Learning how to prove Workshop Week 5

Permutations

Even and Odd Permutations

1. The **sign of a permutation** $\sigma \in S_n$ is defined as

$$\operatorname{sgn} \sigma = \begin{cases} 1 & \text{if } \sigma \text{ is even} \\ -1 & \text{if } \sigma \text{ is odd} \end{cases}$$

   Prove that $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}\sigma \operatorname{sgn}\tau$.

2. **Prove that every r-cycle is a product of $r-1$ transpositions:**

$$(k_1\ k_2 \ldots k_r) = (k_1\ k_2)(k_2\ k_3)\ldots(k_{r-1}\ k_r) = (k_1\ k_r)(k_1\ k_{r-1})\ldots(k_1\ k_2)$$

   Without factorising into transpositions, find the parity of $\sigma = (1\ 3\ 2\ 5\ 6)(2\ 3)(4\ 6\ 5\ 1\ 2)$.